Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/24/2006
PI Name:	Shepherd, Ian Ph.D.		
Project Title:	A Numerical Investigation of Buoyancy-Induced Pressure Effects in Premixed Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Suffix: Assume PhD; referred to as Dr. in NASA research award announcement in 2002 -- rje 2/1/2017		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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	A large body of experimental data has been accumulated with NASA funding under +1g, -1g and micro-g conditions which show the effects of buoyancy on premixed combustion processes. The purpose of this investigation is to model these data numerically and thereby test their adequacy and also to help elucidate the mechanisms by which buoyancy interacts with premixed laminar and turbulent flames. Turbulent combustion processes account for approximately 90% of the power and heat generated worldwide but remain among the most significant unresolved problems in combustion science. In these processes mass and momentum fluxes, chemical kinetics and heat release are coupled in complex ways over a wide range of time and length scales. With the increased computational power now available formerly intractable difficulties can be addressed and the Computational Fluid Dynamics (CFD) codes being developed will become useful design tools for the practical engineer and, to a		

Task Description:	significant extent, obviate the need for often expensive, and time consuming empirical work. However, due to the current incomplete state of knowledge of turbulent combustion processes a close coordination of experimental and computational approaches is necessary for advances to be made. Combustion research and technology is also directed at reducing pollution formation by burning at lean stoichiometries. Under these conditions flame instability, blow-off and flashback become safety concerns. Buoyancy has been observed to have a significant effect on flame stability and on conditions in the far-field burnt gas which, through the pressure field, affect flow both upstream and downstream of the flame. The temperature increase across a typical hydrocarbon flame is large (factors of ~6-8) and so the flow field consists predominately of hot, light products downstream and cold, dense reactants upstream in the reactants. The buoyancy affects the downstream flow by accelerating the products and has a large impact on the boundary conditions of the flow. These large scale effects associated with the interaction of buoyancy with laminar and turbulent flames have yet to be adequately simulated. This is primarily due to the necessity of modeling a large physical domain so as to be insensitive to boundary conditions associated with buoyancy. In this study the near and far field of premixed laminar and turbulent flames will be simulated using a local adaptive mesh refinement (AMR) technique. The advantage of this technique is that it is capable of modeling large domains without sacrificing numerical resolution where it is required. The purpose of this study is, by numerical simulation of premixed flames, to elucidate the mechanisms by which buoyancy effects in the far field can modify premixed flame structure and dynamics. The work ise based on, and attempts to simulate, experimental work previously obtained under NASA funding. Collecting experimental data under microgravity conditions is a difficult and expensive enterprise and so the numerical simulations while aiding in the interpretation of experimental data but are at the same time be able to supplement those measurements. Furthermore, study of the boundary conditions associated with buoyant interactions provides insight into the modeling of the more complex boundaries of, for example, walls and physically confined flows. This is a problem of practical importance as the interaction of buoyancy and walls in forming the boundary conditions of large industrial flames often has to be dealt with in an ad hoc manner. The buoyancy/premixed flame interaction could be considered, therefore, a paradigm for large-scale pressure effects and so the results of this study will be relevant not only for combustion science but also for technological development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion processes account for approximately 90% of the power and heat generated worldwide but remain among the most significant unresolved problems in combustion science. With the increased computational power now available formerly intractable difficulties can be addressed and numerical simulations will become useful design tools for the practical engineer and, to a significant extent, obviate the need for often expensive, and time consuming empirical work.
Task Progress:	The purpose of this investigation is to model numerically a large body of experimental data that has been accumulated under NASA funding at +1g, -1g and micro-g conditions which show the effects of buoyancy on premixed combustion processes. These simulations are designed to elucidate the mechanisms by which buoyancy interacts with premixed laminar and turbulent flames. In this study the near and far fields of premixed laminar and turbulent flames are simulated using a local Adaptive Mesh Refinement (AMR) technique. The advantage of this technique is that it is capable of modeling large spatial domains without sacrificing numerical resolution where it is required. With the increased computational power now available formerly intractable problems can be addressed and the Computational Fluid Dynamics (CFD) codes being developed, such as AMR will become useful design tools for the practical engineer. The AMR numerical code has been adapted for the study of buoyant and microgravity phenomena and has been successfully applied to the case a two dimensional, axisymmetric premixed stoichiometric laminar methane/air Bunsen flame stabilized on a 2.5 cm burner, performed in a large 32 x 64 cm computational domain. 1) Flame tip flicker in premixed conical laminar methane/air flames has been simulated. These simulations, using a reduced chemical scheme, are performed with an inlet flow of 0.73 m/s and equivalence ratio, phi=0.8. Time and phase-averaged velocity fields, centerline axial velocities and the flame flicker frequency (10.2 Hz) show good agreement between the numerical and experimental results. The flame tip oscillation is caused by a competition between the pressure fields associated with the predominately radial motion of the burnt gases near the flame front and the rotating vortex motion. The phase-averaged velocity data provided an excellent means for following the evolution of the flow field but a more detailed investigation of the flicker process was also performed. Two types of vortex evolution were observed as alternate vortices were entrained leading to regular variations in the maximum flame tip height. 2) A study of buoyancy affects under gravity conditions characteristic of lunar and Martian conditions was started. Two aspects were investigated: flame licker frequency and inverse gravity fields. The affect of the reducing buoyant forces is clear from the lowered flicker frequency and flicker amplitude. 3) Work is continuing on the flame flicker phenomenon with a much wider range of initial conditions. Close comparisons with the experimental data already obtained with NASA funding is a central concern of this study. 4) The ‘funnel’ flame, an axiymmetric V-flame for which experimental data is available is also being studied. Of particular interest is the affect of buoyancy on the included flame angle under normal and inverse gravity conditions for which a large experimental data base exists. 5) A study of the ignition of a conical flame under 1g and 0g conditions in an attempt to obtain some insight into the dynamics of a premixed flame during the time, typically 2 secs, of a drop tower experiment. A large amount of experimental data for microgravity combustion has been obtained form drop-tower experiments, specifically from the drop-tower in NASA-Glenn. A concern with these experimental results is the affect of the transition from 1g to 0g and how long the induced stabilities take to damp out. Only after this has happened are reliably microg results obtained. Our numerical approach provides a good means to investigate these phenomena.
Bibliography Type:	Description: (Last Updated: 07/24/2006)
Articles in Peer-reviewed Journals	Gotoda H, Ueda T, Shepherd IG, Cheng RK. "Flame Flickering Frequency on a Rotating Bunsen Burner." Chemical Engineering Science, in press, 2006. , Aug-2006